

RISHOBAN KANDEEPAN

Embedded Firmware Development | Embedded C/C++ | ARM Cortex-M/ESP32 | RTOS & Secure Systems

Colombo, Sri Lanka | rishobankandeepan@gmail.com | +9475 768 19 94

linkedin.com/in/rishoban-kandeepan | Rishoban Kandeepan-Portfolio | github.com/RishobanK

Summary

Final-year Electrical and Electronic Engineering undergraduate with hands-on experience in embedded C/C++ firmware development, microcontroller-based control systems, sensor integration, and hardware-software debugging. Experienced in STM32/ARM Cortex-M bootloader development, firmware validation, flash memory handling, CRC32 integrity checking, and AGV firmware development with PID-based line following, EEPROM storage, and real-time sensor handling. Strengthened embedded software knowledge through certifications and hands-on projects covering Embedded C, BSP development, RTOS, Embedded Linux, ARM Cortex-M/STM32, and secure embedded systems. Interested in reliable firmware development, firmware security, testing, and technical documentation.

Technical Skills

Languages: C, C++, Python, Assembly; basic Java

Firmware & Low-Level: Bare-metal firmware, bootloader design, flash memory programming, linker scripts, vector table relocation, interrupt-driven logic, timers/PWM, ADC acquisition, EEPROM/NVM storage, sensor calibration, state-machine design, serial debugging

Platforms & Interfaces: ESP32, STM32/ARM Cortex-M, ATmega2560, PIC16F877A, Raspberry Pi; UART, I2C, SPI, GPIO, PWM, ADC, TCP/IP

RTOS, Linux & Security Focus: FreeRTOS task scheduling concepts, Embedded Linux GPIO/I2C/UART interaction, firmware validation, CRC32 integrity checking, secure boot concepts, firmware signing, TLS/HTTPS, certificate validation

Tools & PCB Design: Git, GitHub, VS Code, PlatformIO, STM32CubeIDE, STM32CubeMX, ST-Link, OpenOCD/GDB, KiCad, EasyEDA, Proteus, MATLAB

Experience

Electronic Engineering Trainee, Variosystems – Sri Lanka June 2025 – Nov 2025

- Developed embedded firmware improvements for an industrial AGV, integrating load-cell calibration, EEPROM-based data persistence, PID line-following, white-zone detection, adaptive turning, and non-blocking timer logic.
- Performed firmware testing, sensor calibration, route validation, and hardware-software debugging to improve AGV reliability under industrial operating conditions.
- Developed a Raspberry Pi-based PCB alignment verification system using Python, Tkinter, and OpenCV, with conveyor/buzzer outputs triggered by real-time orientation validation.

Engineering Trainee, Colombo Dockyard PLC – Sri Lanka Jan 2023 – April 2023

- Assisted in maintenance and troubleshooting of shipboard electrical and communication systems, supporting operational reliability and safety.
- Contributed to energy-efficiency improvements for welding machines by identifying idle power usage issues and supporting practical optimization work.

Education

Sri Lanka Institute of Information Technology (SLIIT), BSc (Hons) in Electrical & Electronic Engineering June 2021 – Present

- **Relevant Coursework:** Embedded Systems Programming, Digital Design, Electronic Design, Robotics and Control, Internet of Things, Computer Vision and Image Processing

Royal College, Colombo Jan 2012 – Oct 2020

- G.C.E. Advanced Level and Ordinary Level completed

Projects

Secure STM32 Bootloader & Firmware Update System – ARM Cortex-M Firmware Security GitHub

- Developed a custom STM32 Blue Pill bootloader with relocated application execution, flash erase/write handling, update flag mechanism, and firmware update workflow.
- Verified bootloader behavior using LED-coded status indicators, ST-Link breakpoints, watch variables, STM32CubeProgrammer flash memory inspection, and ESP32-C3 UART bridge testing.
- **Tools:** STM32 Blue Pill, Embedded C, STM32CubeIDE, ST-Link, Flash Memory, Linker Script, CRC32, Python

Autonomous Guided Vehicle (AGV) – Embedded Firmware & Sensor Integration GitHub

- Developed embedded firmware for an industrial AGV with load-cell calibration, EEPROM persistence, PID-based line following, white-zone detection, adaptive turning, pushbutton override, OLED feedback, and non-blocking timer logic.
- Performed sensor calibration, route validation, and hardware-software debugging to improve navigation reliability under industrial operating conditions.
- **Tools:** Embedded C/C++, ATmega2560, CZL601 Load Cell, IR Sensor Array, EEPROM, OLED, Motor Control

ESP32 Custom Development Board – PCB Design & Embedded Hardware GitHub

- Designed a custom ESP32-based development board for embedded prototyping, including power regulation, boot/reset circuitry, GPIO breakout, programming interface, and sensor/IO expansion.
- Created schematic, PCB layout, and fabrication files while considering component placement, signal routing, power distribution, and debugging accessibility.
- **Tools:** ESP32, KiCad, PCB Design, Schematic Design, Power Regulation, GPIO, UART/I2C/SPI

Wearable Kyphosis Monitoring System – ESP32 Sensor Acquisition & Feedback GitHub

- Developed an ESP32-based wearable monitoring system using EMG and gyroscope sensors for real-time posture-related data acquisition and feedback control.
- Implemented sensor data reading, noise filtering, serial data output, and vibration feedback using micro DC vibration motors.
- **Tools:** ESP32, Raspberry Pi 5, MyoWare Sensor, MPU6050, Embedded C/C++, Python

Vision-Based PCB Alignment Verification – Raspberry Pi Automation GitHub

- Developed a Raspberry Pi-based PCB alignment verification system using Python, OpenCV, Tkinter, ORB feature matching, and real-time ROI selection.
- Integrated orientation validation with conveyor/buzzer outputs to support automated inspection under rotation, scale variation, and partial occlusion.
- **Tools:** Raspberry Pi 5, Python, OpenCV, Tkinter, GPIO

Certifications & Professional Development

Selected Certifications: Advanced BSP Development with Embedded C; Advanced Architecture in Embedded Software Design; Foundations of Embedded Software Design; Embedded Systems using C

Current Development Focus: FreeRTOS, Embedded Linux, authenticated firmware update, firmware signing, TLS/HTTPS communication, certificate validation, and secure embedded communication.

Volunteering

Robofest 2023 & 2024 – Supported outreach activities for a nationwide robotics competition, including student guidance on line-following robot concepts.

Referees

Dr. Eranga Wijesinghe

Professor, Faculty of Engineering, SLIIT
eranga.w@slit.lk | +9474 153 10 13

Eng. T. Shanmugam

Director, Interprof International
s_thangamuthu@yahoo.com | +9477 066 28 25